

PUJIT BALANTHIRAN

pujitbalanthiran23@gmail.com | linkedin.com/in/pujitbalanthiran | github.com/pujit23 | +91 8310845246

SUMMARY

Computer Science & Engineering Sophomore | Vellore Institute of Technology - AP University

Software Engineering student with hands-on experience building production-grade, full-stack systems powered by AI pipelines, real-time data processing, and cloud-native architecture. Proficient in Java, Python, C++, JavaScript, and React.js, with a strong foundation in Data Structures and Algorithms, Object-Oriented Programming, and Software Engineering principles. Independently conceptualizes, architects, and delivers scalable end-to-end solutions - from graph database models and serverless cloud functions to real-time dashboards and intelligent recommendation engines. A driven problem-solver who bridges theoretical foundations with real-world engineering to build systems that matter.

TECHNICAL SKILLS

Programming - Java, Python, C++, JavaScript, React.js

Frontend - HTML5, CSS3

Backend - Node.js, Express.js

Database - MongoDB, SQL

Tools - Git, Firebase, MATLAB, RStudio, Vivado

AI / Cloud - Google Gemini Pro, Google Speech-to-Text API, WhatsApp Cloud API, GitHub Actions CI/CD

CS Core - Object-Oriented Programming, Data Structures and Algorithms, Cryptography, Problem Solving, Database Management Systems, Software Engineering.

PROJECTS

NeedGraph | AI-Powered Crisis Prediction and Volunteer Coordination Platform

Tech Stack: React, Vite, Tailwind CSS, D3.js, Leaflet.js, Node.js, Google Cloud Functions, Google Gemini Pro, Google Speech-to-Text API, Neo4j, WhatsApp Cloud API, Google Cloud Run, GitHub Actions CI/CD

- Engineered a massively scalable, real-time AI pipeline that ingests WhatsApp voice notes in regional languages via Google Speech-to-Text API, processes unstructured field data through Google Gemini Pro to extract structured severity scores (0-100), need classification, affected population count, and geo-location - delivering actionable intelligence in under 60 seconds end-to-end.
- Architected a graph database model using Neo4j to map and traverse interconnected social risk nodes, implementing a cascade prediction algorithm that detects multi-level crisis chains (food insecurity to school dropout to mental health crisis) and triggers automated coordinator alerts up to 3 weeks before escalation — transforming reactive response into proactive risk management.
- Built and deployed a 13-page low-latency React dashboard featuring a D3.js network graph explorer, Leaflet.js real-time need map, crisis simulator, and live volunteer dispatch system on Google Cloud Run with automated CI/CD via GitHub Actions, designed to support coordination across 10,000+ NGOs operating in India.

CycHigh | Full-Stack Cycling Encyclopedia and Community Platform

Tech Stack: React 18, Vite, Tailwind CSS, Framer Motion, Recharts, Node.js, Express.js, MongoDB Atlas, Mongoose, JWT, bcryptjs, Multer

- Designed a scalable full-stack web application backed by a MongoDB Atlas database of 120 bicycles across 25 brands, with complete technical specifications, INR pricing, maintenance guides, and upgrade recommendations.

- Implemented an intelligent 5-parameter recommendation algorithm scoring user inputs across skill level, terrain type, budget, and riding purpose to surface top 5 matched cycles with percentage scores; engineered a side-by-side comparison engine with automated value-highlighting logic and client-side PDF export.
- Built a secure, scalable REST API backend using Node.js and Express.js with JWT-based authentication, bcryptjs password hashing, Multer-based image upload handling, rate limiting, and a full admin dashboard supporting complete CRUD operations and bulk JSON import for 120+ records.

FindSpace | Secure Campus Peer-to-Peer Marketplace

Tech Stack: React 19, React Router DOM v7, Vite, Vanilla CSS, Lucide React, Firebase Authentication

- Developed a secure, identity-verified peer-to-peer trading platform using React 19 and Firebase Authentication, enforcing campus-exclusive access through OTP-based phone verification — ensuring zero unauthorized transactions.
- Engineered a multi-context state management architecture using React Context API (AuthProvider, CartProvider, ProductProvider, Theme Context) to manage authentication state, cart logic, real-time product listings, and theme switching across 8+ application pages.
- Delivered a fully custom, mobile-responsive CSS design system leveraging CSS variables, glassmorphism, dynamic gradients, and micro-animations — with ARIA-compliant focus trapping across all modal components for accessibility compliance.

HACKATHONS AND COMPETITIONS

- **Google TechSprint** — Top 10 Finalist out of all participating teams; built and presented a functional technical solution under time constraints.
- **Google DevSprint** — Participated and collaborated in a fast-paced engineering sprint organized by Google Developer ecosystem.
- **Google Solutions Challenge** — Developed a socially impactful technology solution addressing UN Sustainable Development Goals.
- **Smart India Hackathon (SIH)** — Participated in India's largest national-level hackathon; proposed and prototyped an engineering solution for a real government problem statement.

CERTIFICATIONS

- **HackerRank SQL (Basic)** — HackerRank
- **HackerRank SQL (Intermediate)** — HackerRank
- **SQL** — Kaggle
- **MATLAB Onramp** — MathWorks

CLUBS AND COMMUNITY

- **Microsoft Student Chapter, VIT-AP** — Active member contributing to technical workshops, peer learning, and collaborative engineering initiatives.
- **Computer Society of India (CSI), VIT-AP** — Member engaging in technical events, seminars, and computer science community activities.
- **Google Developer Group (GDG), VIT-AP** — Active member participating in Google-ecosystem developer events, tech talks, and hands-on coding sessions.

EDUCATION

Vellore Institute of Technology – AP University

2024 - 2028

Bachelors of Technology in Computer Science and Engineering

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Digital Image Processing, Assembly Language Programming, Software Engineering, Database Management Systems, Cryptography.